

TULSA JONES, OK, USA

WMO: 723564

Lat: 36.039N

Lon: 95.984W

Elev: 195

StdP: 99.01

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 53908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.7	(c) -7.4	(d) -16.6	(e) 0.9	(f) -7.1	(g) -14.0	(h) 1.1	(i) -4.7	(j) 9.5	(k) 12.2	(l) 8.7	(m) 12.5	(n) 2.5	(o) 20	(p) 0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	38.1	24.4	36.8	24.8	35.2	24.7	26.5	34.2	25.8	33.3	25.4	32.8	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.7	29.8	23.8	19.1	29.4	23.1	18.3	28.9	83.4	34.1	80.9	33.5	78.6	32.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.8	(b) 7.9	(c) 7.2	(e) -14.6	(f) 40.1	(g) 3.0	(h) 2.2	(i) -16.8	(j) 41.6	(k) -18.5	(l) 42.9	(m) -20.2	(n) 44.1	(o) -22.4	(p) 45.7		
(a) 8.8	(b) 7.9	(c) 7.2	(e) -15.2	(f) 27.5	(g) 2.8	(h) 1.1	(i) -17.3	(j) 28.3	(k) -18.9	(l) 28.9	(m) -20.5	(n) 29.5	(o) -22.5	(p) 30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	3.6	5.9	11.0	16.2	20.9	25.8	28.2	27.8	23.4	16.8	10.4	4.7
	DBStd	9.86	5.66	6.01	5.75	4.73	4.10	2.97	2.71	2.89	3.90	4.87	5.28	5.38
	HDD10.0	656	209	141	56	7	0	0	0	0	0	6	59	178
	HDD18.3	1919	456	351	235	95	23	1	0	0	6	89	242	423
	CDD10.0	2950	12	25	87	192	339	474	566	553	403	215	70	14
	CDD18.3	1171	0	1	8	30	103	225	307	294	158	40	3	1
	CDH23.3	13422	2	12	76	306	1006	2544	3861	3532	1641	401	42	1
	CDH26.7	6583	0	2	13	72	343	1202	2130	1942	754	120	5	0
Wind	WSAvg	2.9	2.9	3.2	3.5	3.6	3.2	3.1	2.5	2.3	2.4	2.6	2.8	2.8
Precipitation	PrecAvg	1007	38	43	78	110	138	126	88	79	103	81	67	58
	PrecMax	1458	97	102	196	250	332	258	234	257	246	210	182	205
	PrecMin	687	3	3	21	38	23	27	7	1	28	28	0	2
	PrecStd	198	26	26	43	51	71	61	56	56	53	48	52	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	25.0	28.3	31.3	33.7	37.4	40.2	41.3	38.7	32.4	27.3	22.4
		MCWB	13.2	15.1	17.5	19.3	23.1	24.5	23.8	24.1	23.5	22.2	17.9	16.4
	2%	DB	19.1	22.0	25.5	28.6	32.1	35.4	38.0	38.4	34.9	29.9	24.2	19.2
		MCWB	11.9	14.4	16.2	19.1	22.4	24.8	24.4	24.5	23.6	20.8	16.1	13.3
	5%	DB	16.8	19.1	23.0	26.9	30.4	33.9	36.7	37.0	33.0	27.6	22.1	16.9
		MCWB	11.1	12.9	15.4	18.0	21.9	24.3	25.0	24.8	23.2	19.3	15.5	11.8
	10%	DB	13.3	16.7	21.1	24.9	28.6	32.5	35.1	35.2	31.2	25.4	19.6	13.9
		MCWB	8.2	11.1	14.6	17.4	21.3	23.8	24.9	24.7	22.5	18.5	14.0	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	18.0	19.5	21.9	25.0	26.9	27.4	27.2	25.6	23.5	19.8	18.0
		MCDB	19.5	22.2	25.9	28.0	31.2	34.1	34.4	35.1	32.8	29.7	24.3	21.2
	2%	WB	14.0	16.2	18.1	20.7	23.9	25.9	26.5	26.3	24.8	22.5	18.0	15.3
		MCDB	17.4	20.3	23.4	26.1	29.5	32.8	34.2	34.4	32.0	27.6	21.7	18.1
	5%	WB	11.1	14.1	16.8	19.8	23.1	25.3	25.9	25.7	24.1	21.4	16.6	12.6
		MCDB	15.7	18.3	21.6	24.6	28.5	31.8	33.4	33.6	31.0	25.6	20.7	15.1
	10%	WB	8.4	11.1	15.4	18.7	22.2	24.5	25.5	25.1	23.4	19.9	15.0	10.0
		MCDB	12.8	16.0	20.2	23.4	27.4	30.8	32.9	32.8	29.5	24.0	19.0	13.9
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.3	12.5	12.5	11.4	11.0	11.8	12.4	12.5	12.8	12.9	11.6
		MCDBR	17.7	16.6	15.7	15.1	13.1	12.4	14.1	15.2	14.4	15.0	16.0	16.1
		MCWBR	11.3	10.4	8.6	7.7	5.5	4.4	4.1	4.4	4.9	7.1	9.3	10.6
	5% WB	MCDBR	15.0	14.3	12.3	12.0	10.9	11.0	12.0	12.6	12.1	11.5	13.2	12.5
		MCWBR	10.7	10.4	7.6	7.4	5.5	4.6	4.4	4.5	4.6	6.4	9.3	10.1
Clear-Sky Solar Irradiance	taub	0.302	0.309	0.352	0.387	0.425	0.433	0.442	0.443	0.395	0.352	0.318	0.298	
	taud	2.494	2.489	2.398	2.318	2.274	2.281	2.298	2.258	2.365	2.472	2.510	2.518	
	Ebn at Noon	900	935	915	896	863	852	841	833	859	873	873	882	
	Edh at Noon	82	92	111	127	135	134	130	133	114	93	80	75	
All-Sky Solar Radiation	RadAvg	2.58	3.16	4.16	5.19	5.71	6.55	6.60	5.98	4.99	3.74	2.74	2.19	
	RadStd	0.26	0.47	0.25	0.33	0.54	0.64	0.38	0.39	0.42	0.47	0.34	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page